
Chapter 1: Installation Prerequisites for Linux

This document describes the steps necessary to correctly install and configure all Ansys, Inc. products on Linux platforms for Release 2026 R1. To view a complete list of the products available, see [Applications Included with Each Product](#) (p. ?).

Notes

Not all products may be available on all Linux platforms. See the remainder of this document for detailed information on which products are available on which platforms.

At Release 2026 R1, Ansys BladeGen, Vista CCD, Vista CPD, Vista RTD, and Vista AFD are not supported on Linux platforms.

Ansys Workbench is installed by default as product components to most Ansys, Inc. products. Ansys Workbench is not installed as part of the products under Ansys Additional Tools. Ansys Workbench includes the following applications:

- DesignModeler
- Design Exploration
- Meshing
- Remote Solve Manager
- Fluent Meshing
- EKM Client

Important Notice

If you wish to run multiple releases of Ansys, Inc. software, you **MUST** install them chronologically (that is, Release 2025 R2 followed by Release 2026 R1). If you install an earlier release after installing Release 2026 R1, you will encounter licensing issues that may prevent you from running any products/releases. If you need to install an earlier release after you have already installed Release 2026 R1, you **MUST** un-install Release 2026 R1, then re-install the releases in order.

Summary of New and Changed Features

The following features are new or changed at Release 2026 R1. Review these items carefully.

- For this release, the Ansys installation program has added the following products:
 - Connect
 - Behavior Execution Engine

- Electronics
 - PerceiveEM
- Embedded Software
 - Scade One
 - SCADE Rapid Prototyper
- Need for PAC file, and its URL, are automatically detected from operating system settings.

1.1. System Prerequisites

Ansys, Inc. Release 2026 R1 products are supported on the Linux platforms and operating system levels listed in the following tables. Patches listed may be superseded by newer versions; see your vendor for the most current version. See the individual Platform Certification Specifics in this guide for more information on specific platform requirements. For up-to-date information on hardware platforms or operating systems that have been certified, go to the [Platform Support section of the Ansys Website](#).

Table 1.1: Supported Linux Platforms

Platform	Processor	Operating System	Platform architecture (directory name)	Availability
Linux x64	EM64T/Opteron 64	Red Hat Enterprise Linux 8.8 & 8.10, Red Hat Enterprise Linux 9.4 & 9.6, SUSE Linux Enterprise Server and Desktop 15 SP5 - SP7, Ubuntu LTS Desktop and Server 22.04 & 24.04, Rocky Linux 8.10, 9.4 & 9.6	linux64	Download / USB
For detailed and current support information, see the Platform Support section of the Ansys Website .				

Supported Platforms for High Performance Computing

See the discussions on [Configuring Distributed Ansys](#) and [Configuring Ansys CFX Parallel](#) for detailed information on supported platforms for distributed computing.

Supported Products By Platform

For a complete list of products supported by each platform, see the **Platform Support by Application** PDF on the [Platform Support section of the Ansys Website](#).

1.1.1. CAD Support

The following CAD and auxiliary programs are supported on the indicated products and platforms. Products are:

A = Mechanical APDL

W = Ansys Workbench

I = Ansys ICEM CFD (standalone and Ansys Workbench Readers)

Table 1.2: CAD Plugin Support by Platform

	Linux x64
CATIA 4.2.4	A, I
CATIA V5 R2023	W, I
CATIA V6 R2024.x	W, I
Parasolid 35.1	A, W, I
ACIS 2023	A, W, I 1
NX 22065, 2212 5, 2306 5	A, W, I
STEP AP203, AP214, AP 242	A, W, I
IGES 4.0, 5.2, 5.3	A 2, W, I 3
GEMS	I
Rhinoceros	I4

1. For Ansys ICEM CFD standalone, ACIS 18.0.1 is the supported version for all platforms.
2. MAPDL supports 5.1 by default, but 5.2 is also supported if the IOPTN command is used.
3. IGES Versions 4.0, 5.2, and 5.3 are supported.
4. Support limited to Rhinoceros 3 and earlier.
5. Support limited to SuSE Linux Enterprise 15.

Table 1.3: CAD Reader Support by Platform

	Linux x64
CATIA 4.Reader	A, I
CATIA 5 Reader	W, I
CATIA 6 Reader	W, I
NX Reader	A, W, I
AutoCAD Reader	W, I
Autodesk Inventor Reader	W, I
Creo Parametric Reader	A, W, I
SOLIDWORKS Reader	W, I
SOLIDWORKS Reader	W, I

1.2. Disk Space and Memory Requirements

You will need the disk space shown here for each product for installation and proper functioning. The numbers listed here are the maximum amount of disk space you will need. Depending on the options selected for each product and shared product features, you may require less disk space when installing multiple products.

Product	Disk Space (GB)
Ansys Additive	41.0
Ansys Aqwa	32.2
Ansys Autodyn	32.3
Ansys AVxcelerate Sensors	41.8
Ansys AVxcelerate Sensors Library for CarMaker	17.2
BladeModeler	29.7
Ansys CFD Post only	34.8
Ansys CFX (includes CFD-Post)	37.6
Ansys Chemkin	35.5
Ansys Customization Files for User-Programmable Features	32.1
Ansys Dynamic Reporting	39.1
Ansys EnSight	51.4
Ansys Electromagnetics Suite	61.6
Ansys Electromagnetics RSM	12.4
Ansys FENSAP-ICE	36.5
Ansys Fluent (includes CFD-Post)	78.7
Ansys Forte (includes EnSight)	42.9
Ansys Geometry Interfaces	21.9
Ansys ICEM CFD	32.6
Ansys Lumerical	15.9
Ansys Lumerical Photonic Verilog-A	12.8
Lumerical-RPM	13.4
Ansys LS-DYNA	35.3
Ansys Mechanical Products	38..8
Ansys ModelCenter Remote Execution	13.5
Ansys Model Fuel Library (Encrypted)	35.4
Ansys Motion	33.6
Ansys optiSLang Core Headless	17.6
Ansys optiSLang Core UI	18.6
Ansys optiSLang AI	20.8
Ansys optiSLang App Generation	18.4
Ansys Polyflow Classic (includes CFD-Post)	71.0
Ansys Reaction Workbench	35.4
Ansys Remote Solve Manager	16.2
Ansys Rocky	25.8
Ansys ScadeOne	12.8
Ansys Sherlock	38.7
Ansys Speos HPC	13.3

Product	Disk Space (GB)
AnsysSpeos RPC	12.7
Ansys TurboGrid	26.7
Ansys TwinAI	18.6
Disk space required to install all products:	203.0

Memory Requirements

You must have a minimum of 8 GB of memory to run product installations, 16 or 32 GB of memory is recommended.

1.3. Requirements for the GPU Accelerator in the Mechanical APDL Application

Your system must meet the following requirements to use the GPU accelerator capability in Mechanical APDL. For information on the most recently tested GPU cards, see the **GPU Accelerator Capabilities** PDF on the [Platform Support section of the Ansys Website](#).

- The machine(s) being used for the simulation must contain at least one GPU card.
- A minimum 16GB of on-card memory is recommended in order to achieve meaningful acceleration in simulations that can use the GPU card.
- To achieve optimal performance, only GPU cards with significant double precision performance (FP64) are recommended for use with the sparse direct solver and eigensolvers based on the sparse solver (for example, Block Lanczos or subspace). The following cards are recommended:

Card ^[b]	GPU Information ^[a]				Recommended Equation Solvers		
	Release Year	FP64 and FP32 Performance (TFLOPS)	Memory Capacity (GB)	Memory Bandwidth (GB/sec)	Sparse (Direct)	Iterative (PCG, etc.)	Mixed
Data Center Cards							
AMD Instinct MI355X ^[c]	2025	78.6/157.3	288 GB	8,000 GB/s	Y	Y	Y
AMD Instinct MI325X	2024	81.7/163.4	256 GB	6,000 GB/s	Y	Y	Y
AMD Instinct MI300X	2023	81.7/163.4	192 GB	5,300 GB/s	Y	Y	Y
AMD Instinct MI210	2022	22.6/45.3	64 GB	1,600 GB/s	Y	Y	Y
AMD Instinct MI250, MI250X	2021	45.3/90.5	128 GB	3,200 GB/s	Y	Y	Y
AMD Instinct MI250X	2021	47.9/95.7	128 GB	3,200 GB/s	Y	Y	Y

NVIDIA B300	2025	75/1.2	270 GB	7,700 GB/s	Y	Y	Y
NVIDIA B200	2025	80/40	186 GB	7,700 GB/s	Y	Y	Y
NVIDIA B100	2025	___[d]	___[d]	___[d]	Y	Y	Y
NVIDIA H200	2024	34/67	141 GB	4,800 GB/s	Y	Y	Y
NVIDIA H100	2022	30/60	94 GB	3,900 GB/s	Y	Y	Y
NVIDIA L40	2022	1.4/90	48 GB	864 GB/s	N	Y	Y
NVIDIA A30	2021	5.2/10.3	24 GB	933 GB/s	Y	Y	Y
NVIDIA A100	2020	9.7/19.5	80 GB	1,940 GB/s	Y	Y	Y
NVIDIA A40	2020	0.58/37	48 GB	696 GB/s	N	Y	Y
Professional Workstation Cards							
AMD Radeon AI PRO R9700	2025	1.5/47.8	32 GB	640 GB/s	N	Y	Y
AMD Radeon PRO W7900	2023	1.9/61	48 GB	864 GB/s	N	Y	Y
AMD Radeon PRO W7800	2023	1.9/61	48 GB	864 GB/s	N	Y	Y
NVIDIA RTX PRO 6000	2025	1.9/126	96 GB	1,340 GB/s	N	Y	Y
NVIDIA RTX PRO 5000	2025	1/66.5	48 GB	1,790 GB/s	N	Y	Y
NVIDIA A800	2024	9.7/19.5	40 GB	1,500 GB/s	Y	Y	Y
NVIDIA RTX 5880 Ada	2024	1.08/69	48 GB	864 GB/s	N	Y	Y
NVIDIA RTX 5000 Ada	2023	1.0/65	32 GB	576 GB/s	N	Y	Y
NVIDIA RTX 4500 Ada	2023	0.6/39	20 GB	640 GB/s	N	Y	Y
NVIDIA RTX 6000 Ada	2022	1.5/91	48 GB	960 GB/s	N	Y	Y
NVIDIA RTX A5500	2022	0.5/34	24 GB	768 GB/s	N	Y	Y
NVIDIA A16	2021	0.1/4.5	16 GB	200 GB/s	N	Y	Y
NVIDIA A10	2021	1/31	24 GB	600 GB/s	N	Y	Y
NVIDIA RTX A5000	2021	0.5/27	24 GB	768 GB/s	N	Y	Y
NVIDIA RTX A4500	2021	0.3/23	24 GB	640 GB/s	N	Y	Y
NVIDIA RTX A4000	2021	0.3/19	16 GB	448 GB/s	N	Y	Y
NVIDIA RTX A6000	2020	0.6/38	48 GB	768 GB/s	N	Y	Y
Consumer Cards ^[e]							
NVIDIA GeForce RTX 5090	2025	1.6/104	32 GB	1790 GB/s	N	Y	Y
NVIDIA GeForce RTX 5080	2025	0.8/56	16 GB	960 GB/s	N	Y	Y

NVIDIA GeForce RTX 5070 Ti	2025	0.7/44	16 GB	896 GB/s	N	Y	Y
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- [a] The hardware specifications in this table are based on publicly available product data-sheets and vendor documentation. Specifications may change or contain inaccuracies. Verify all details with the official manufacturer resources before making decisions.
- [b] GPUs that are unreleased at the time of this software release are not listed in this table. These GPUs may still be compatible with this release, but their performance and functionality have not been validated.
- [c] Hardware specifications for AMD Instinct MI355X are sourced from the official AMD product datasheet. For details, see <https://www.amd.com/content/dam/amd/en/documents/instinct-tech-docs/product-briefs/amd-instinct-mi355x-gpu-brochure.pdf>.
- [d] The vendor has not declared the specification for this entry.
- [e] Consumer GPUs with 16 GB of memory offer limited performance for large-scale workloads. Their fixed memory capacity restricts acceleration compared to workstation or data center GPUs with higher memory configurations.

- For NVIDIA GPU cards, note the following:

1. The Mechanical APDL installation provides the necessary CUDA 13.0.2 libraries to accelerate the FEA solution. No separate installation of CUDA is required.
2. The GPU driver is not installed by Mechanical APDL. The driver version must be 580.95.05 or newer.

- For AMD GPU cards, note the following:

1. The Mechanical APDL installation provides the necessary HIP/ROCm 7.0.2 libraries to accelerate the FEA solution. No separate installation of HIP/ROCm is required. Note that your machine must have at least SUSE Linux Enterprise Server 15 SP5 to use the ROCm libraries.
2. The GPU driver is not installed by Mechanical APDL. The driver version must be 7.0.2 or newer. Some instructions are given below.

- the following AMD link lists their .

- To install the AMD driver, you must perform the three-step procedure summarized below and described in detail with specific Linux commands for different operating systems on the AMD page: .

- Download the "amdgpu-install" script to your system using the appropriate commands for your operating system. Note that the version of amdgpu-install must come from the ROCm 7.0.2 page, and you only need to download the installer script (the full ROCm package is not needed).

- Install the "kernel-mode driver" with the following command:

```
sudo amdgpu-install --usecase=dkms
```

- Install the "libdrm-amdgpu" library to your operating system using the appropriate commands.

- Add your user name (*LOGNAME*) to the list of users in the video group through the following command.

```
sudo usermod -a -G video $LOGNAME
```

- To utilize a GPU device that is not on the recommended list of cards, set the following environment variable:

`ANSGPU_OVERRIDE=1`

This is most beneficial when you wish to run on newer GPUs that were not available at the time of release of this version of the Ansys program. If you choose to use this environment variable, you should ensure that the GPU device that you wish to use is sufficiently powerful, in terms of both double-precision compute power and on-card memory, to achieve meaningful acceleration for your simulation. Using this environment variable with an underpowered CPU may actually decelerate your simulation.

Support for AMD GPUs released after a specific version of the Mechanical APDL application is limited due to incompatibility between required software libraries and the GPU hardware. To check which GPU architectures are supported, refer to the AMD compatibility matrix version. You can find the GPU architecture in the device's specification sheet.

- If you have issues setting up the GPU drivers, see [Troubleshooting in the *Parallel Processing Guide*](#).
- For details on HPC licensing, see [HPC Licensing in the *Parallel Processing Guide*](#).
- For details on GPU benchmarks, see [Ansys Mechanical Benchmarks](#).
- For more information about using GPUs with the Mechanical APDL application, see the following resources on the Ansys website and innovation space:
 - [What GPU Should I Use – Mechanical APDL Application](#)
 - [Enhancing Mechanical APDL™ performance with GPUs](#)

1.4. Additional Hardware and Software Requirements

- Based upon your platform version and installed products you may need to install additional Linux libraries and packages. For a list of libraries and packages by product, see [Installation Troubleshooting - Mechanical APDL \(p. ?\)](#).
- Intel 64 / AMD64 system with a supported operating system version installed
- 8 GB of RAM
- 128 GB free on the hard drive is recommended
- A current default web browser must be installed to view the **Ansys internet-based help documentation**.

For a complete list of supported browsers, see the **Browser Support** PDF on the [Platform Support section of the Ansys Website](#).

Note that Google Chrome does not support the Linux platforms that Ansys supports.

For a local copy of the product documentation on your system, download an installable version from the Ansys Download Center. Instructions for installing the Local Help are included with the download package.

- TCP/IP for the license manager (see the [Ansys Licensing Guide](#) for more information on TCP/IP)
- Approximately twice as much swap space as memory. The amount of memory swap space on the system may limit the size of the model that can be created and/or solved.
- Discrete graphics card with the latest drivers and compatible with the supported operating systems. For full functionality, use of a recent NVIDIA Quadro or AMD FirePro card with at least 1 GB of discrete video memory and supporting OpenGL version 4.5 or above. Intel Iris Pro Integrated graphics has also been used successfully.

For information on the most recently tested graphics cards, see the **Graphics Cards Tested** PDF on the [Platform Support section of the Ansys Website](#).

Note:

- **Vendor Legacy Support:** To be compatible, legacy GL 4.5 compliant cards should still be within the official legacy support period of their respective graphics vendor, and must still be receiving regular driver updates. Visit the appropriate vendor websites to review legacy hardware lists.
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- A minimum screen resolution of: 1024 x 768 (4:3 aspect ratio), 1366 x 768 (16:9 aspect ratio) or 1280 x 800 (16:10 aspect ratio) with minimum 24 bit color. A higher screen resolution such as 1920x1080 (16:9) or 1920x1200 (16:10) is strongly recommended for most applications.

Note:

- A number of cosmetic display issues have been reported when running the installation program on Linux platforms using AMD graphics cards. These issues do not affect the installation functionality.
 - **Ultra High Definition (4K) Graphics Cards:** Use of Ultra High Definition (4K) graphics cards may cause a number of cosmetic display issues (including enlarged or reduced text and incorrect positioning of option labels). These issues do not affect the functionality of the installation program.
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- X11, OpenGL graphics libraries
- Mesa-libGL (OpenGL) is required to run data-integrated Ansys Workbench applications such as Mechanical.
- For most applications, a three-button mouse is required to access all available functionality.
- PDF reader software is required to read the installation guides and other user documentation.
- Running Ansys Fluent or Ansys CFX on Cray systems requires a minimum of Cray Linux Environment 6.0 update 03 (based on SUSE Linux Enterprise Server 12).

- For Ansys CFX on Cray systems, in order to use the command-line application `cfx5mondata`, you must ensure that library `libsmime3.so` is installed.

1.4.1. AVxcelerate Sensors Software Requirements

For AVxcelerate Sensors, additional hardware and software requirements are listed in [AVxcelerate Sensors Installation Guide](#).

1.4.2. Speos HPC Software Requirements

Speos software offers the ability to run simulation jobs on a Linux cluster using the GPU Cores of the machines. Speos HPC requires a minimum NVIDIA A5000 graphics adapter with a NVIDIA driver 580.95.05.

To correctly run GPU simulation jobs on Linux clusters, Speos HPC requires the `libnvoptix.so` library to be installed for NVIDIA OptiX support.

- Make sure to perform the full installation of the NVIDIA Driver. Otherwise the `libnvoptix.so` library will be missing and the jobs will fail and return the following error:

```
Running the simulation failed!!!
```

```
CUDA error
```

```
CUDA error description : 'Library not found'
```

- If the NVIDIA Driver is already installed, verify that the `libnvoptix.so` library is available on the system:

```
$ ldconfig -p | grep libnvoptix.so
```

```
libnvoptix.so.1 (libc6,x86-64) => /lib64/libnvoptix.so.1
```

- If the `libnvoptix.so` library is not available, reinstall the full driver or manually install the missing component.